Contents

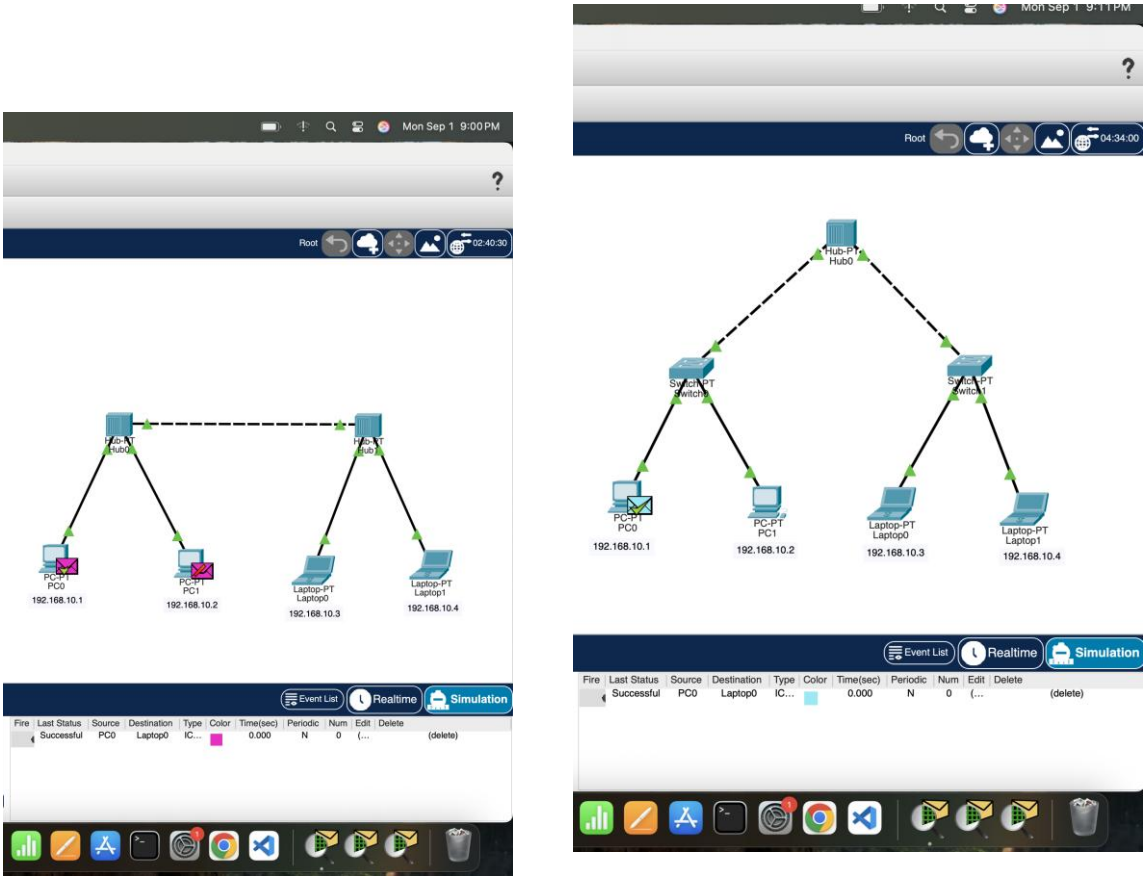
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Task :1-Implementation of LAN

Objective: To know the implementation of Local Area Network

Theory:
A local area network (LAN) is a collection of device connection together in one physical location, such as a building, office, or home. A LAN can be smaller or large, ranging from a home network with thousands of users to an enterprise network with thousands of users and devices in an office or school. A LAN comprises cables, access points switches, routers, and other components that enable devices to connect to internal servers, web servers, and other LANS via wide area networks.

Implementation:
In the packet tracer, computers are connected to a switch and the switch is connected to a router through wires. IP Address is assigned to the router and the computers. Computer must use the IP Address of the router as their default gateway. After assigning IP addresses and gateways the computer should be able to communicate with the router and also with each other. Each computer must be assigned with a unique IP Address.



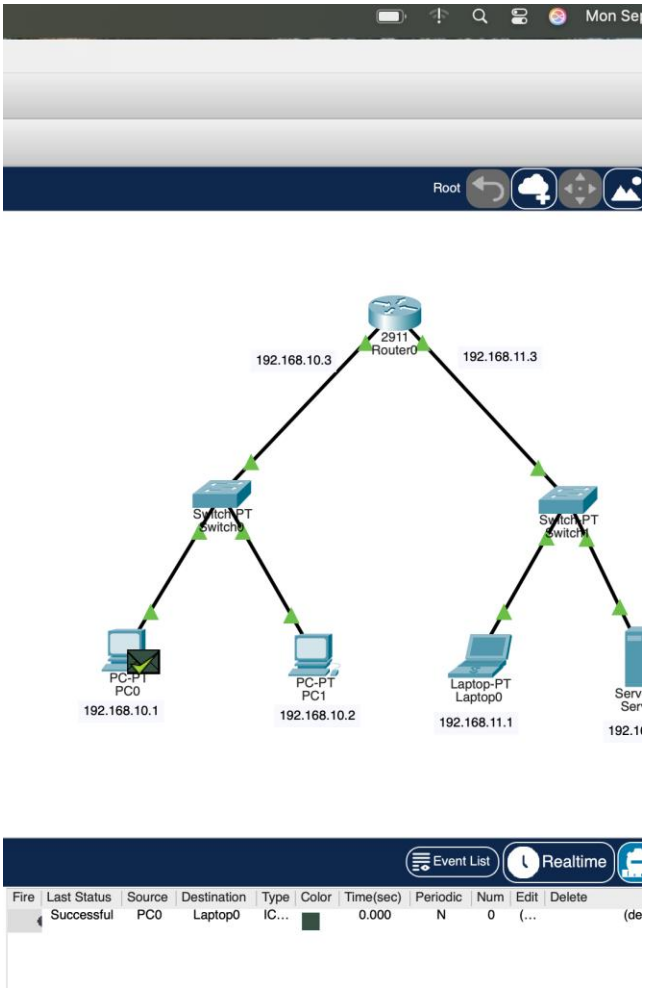

Root
2911
Router0
192.168.10.3
192.168.11.3
PC-PT
PC0
192.168.10.1
PC-PT
PC1
192.168.10.2
Laptop-PT
Laptop0
192.168.11.1
Event List
Realtime
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Successful
PC0
Laptop0
0.000
N
0

Fig: LAN

Task:2 Implementation of DHCP

Objective: To know about implementation of Dynamic Host Configuration Protocol

Theory:
DHCP is a network protocol used on IP network where a DHCP server automatically assigns an IP address and other information to each host on the network so they can communication efficiently with other endpoints. In addition to the IP address, DHCP also assigns the subnet mask, default gateways address, domain name server (DNS) address and other pertinent configuration parameter.

Implementation:
Step1: Set a router with one server and set a desktop as required.
Step2: Enable ports of router and set an IP address for both server and router.
Step3: Go to services on the server and on the services and assign the IP address of the router in the server and set the start IP address.
Step4: Enable the DHCP in the IP configuration section of the desktop.

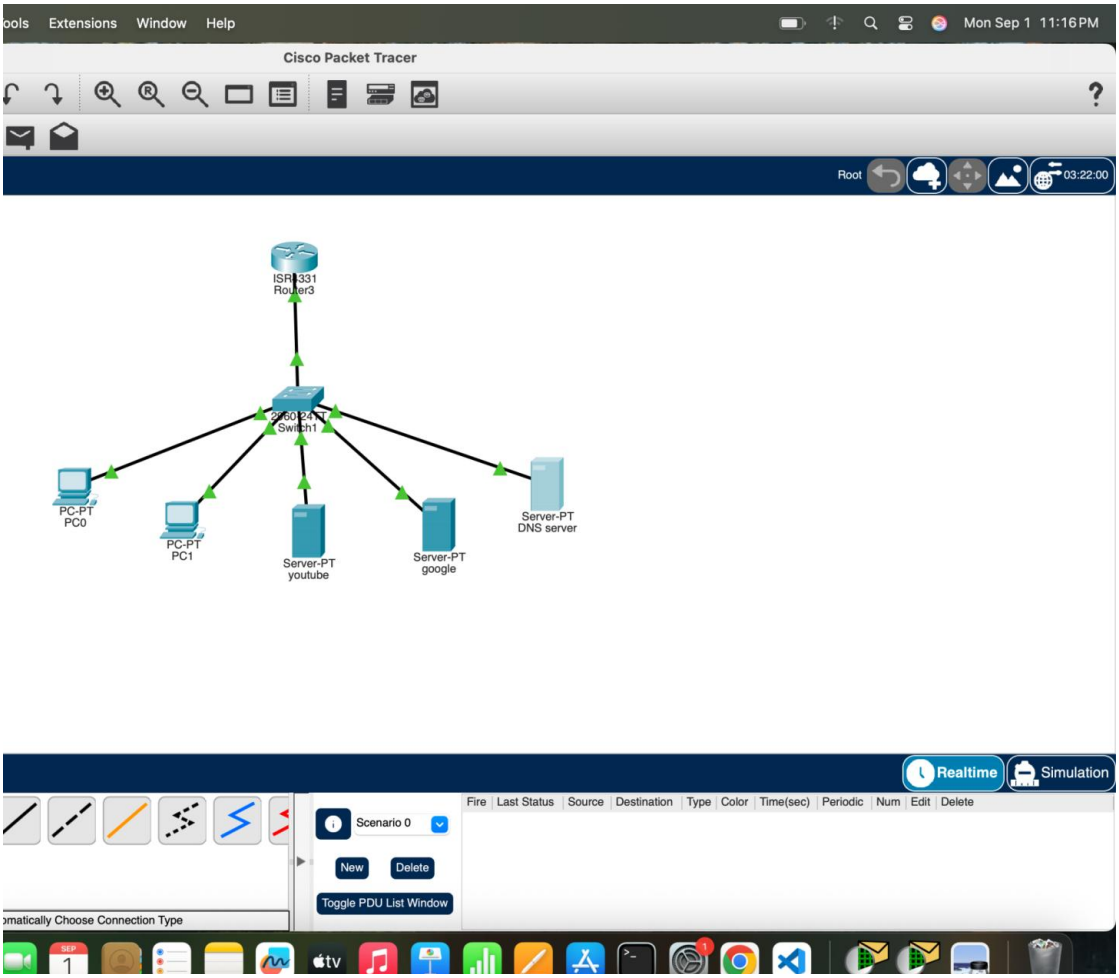


Fig: Server Configuration (DHCP)

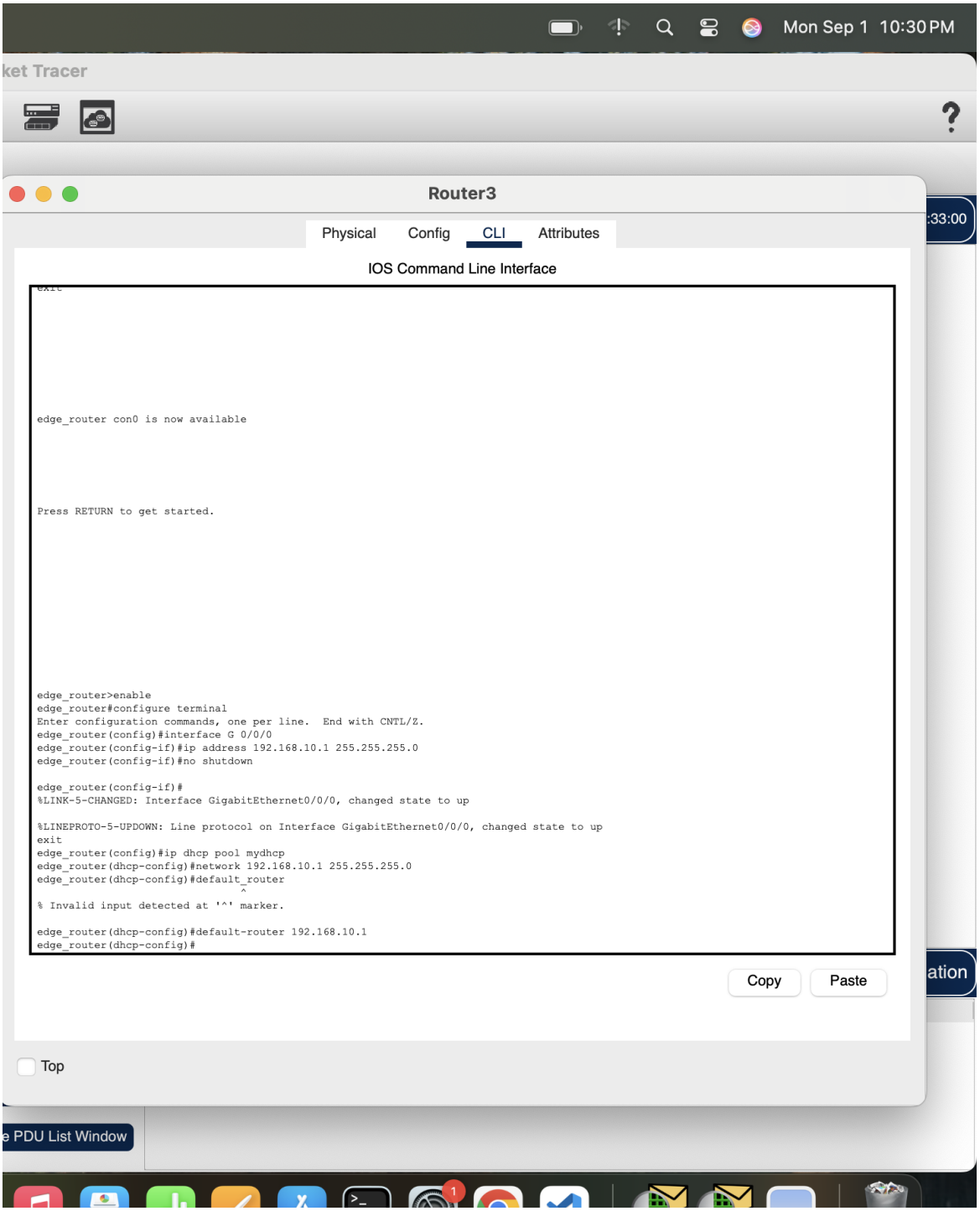
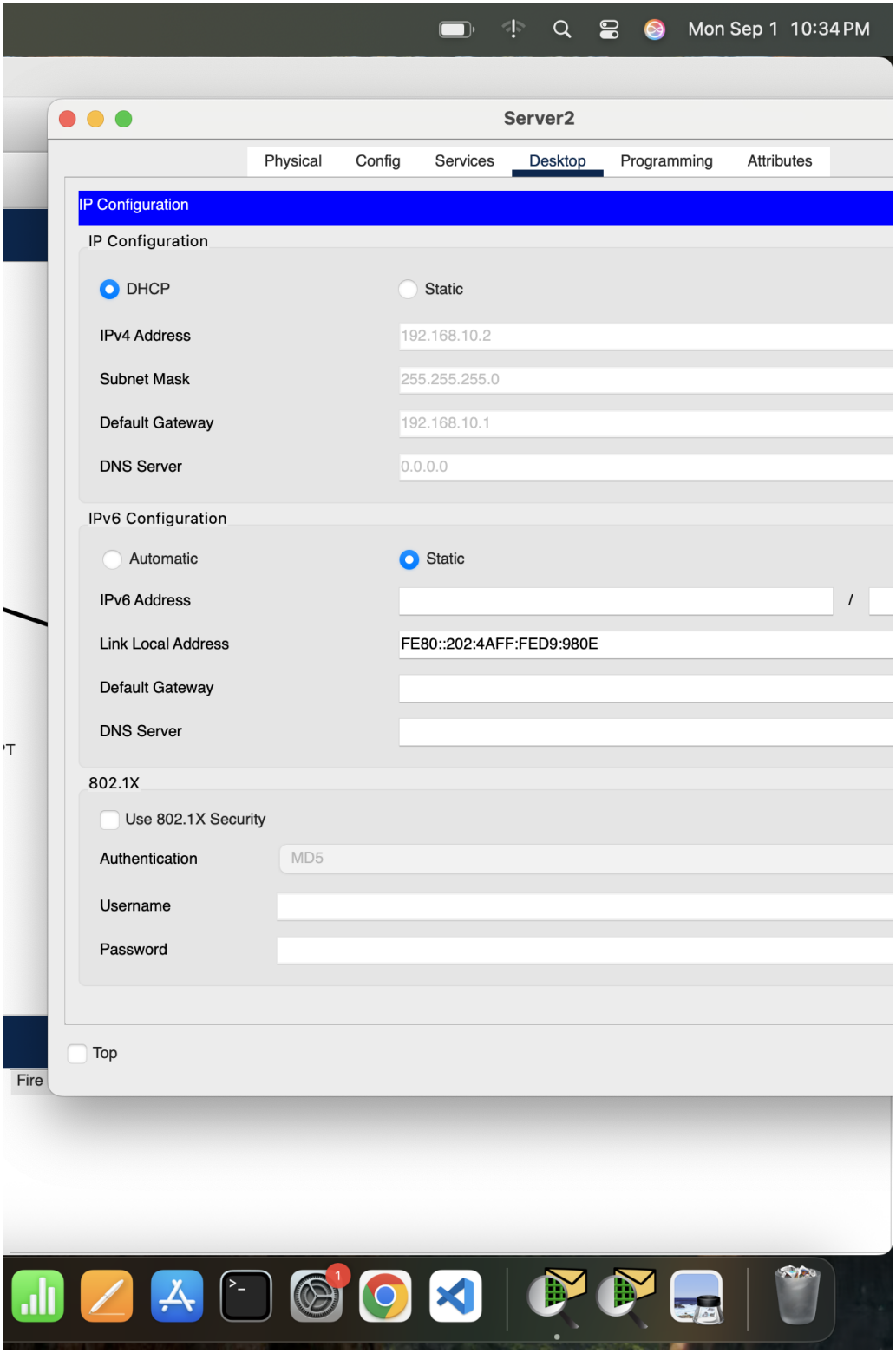


Fig: IP Assignment Dynamically (DHCP)



DHCP server IP assignment

Conclusion:
Hence, DHCP Configuration is implemented using packet tracer.

Task 3: Implementation of DNS

Objective:
To understanding the function and important od DNS service in a network environment.
Configuration a DNS server to manage domain name resolution

DNS server
The purpose of a DNS server is to translate what a user types into their browser into something a computer can understand and use to locate a website. In other words, its purpose is to convert a domain name such as www.example.com into an IP address such as 71.232.101.120.
Thanks to DNS servers, people don't have to memorize complex IP addresses like 216.58.217.206, which is Google's IP address. They just have to memorize www.google.com.This translation process — formally known as DNS resolution — requires multiple hardware components. The most important is known as the primary DNS server.
Configuring a DNS server
To configure a DNS server. We follow the steps below:

Create a network with end devices, switch, and DNS server.

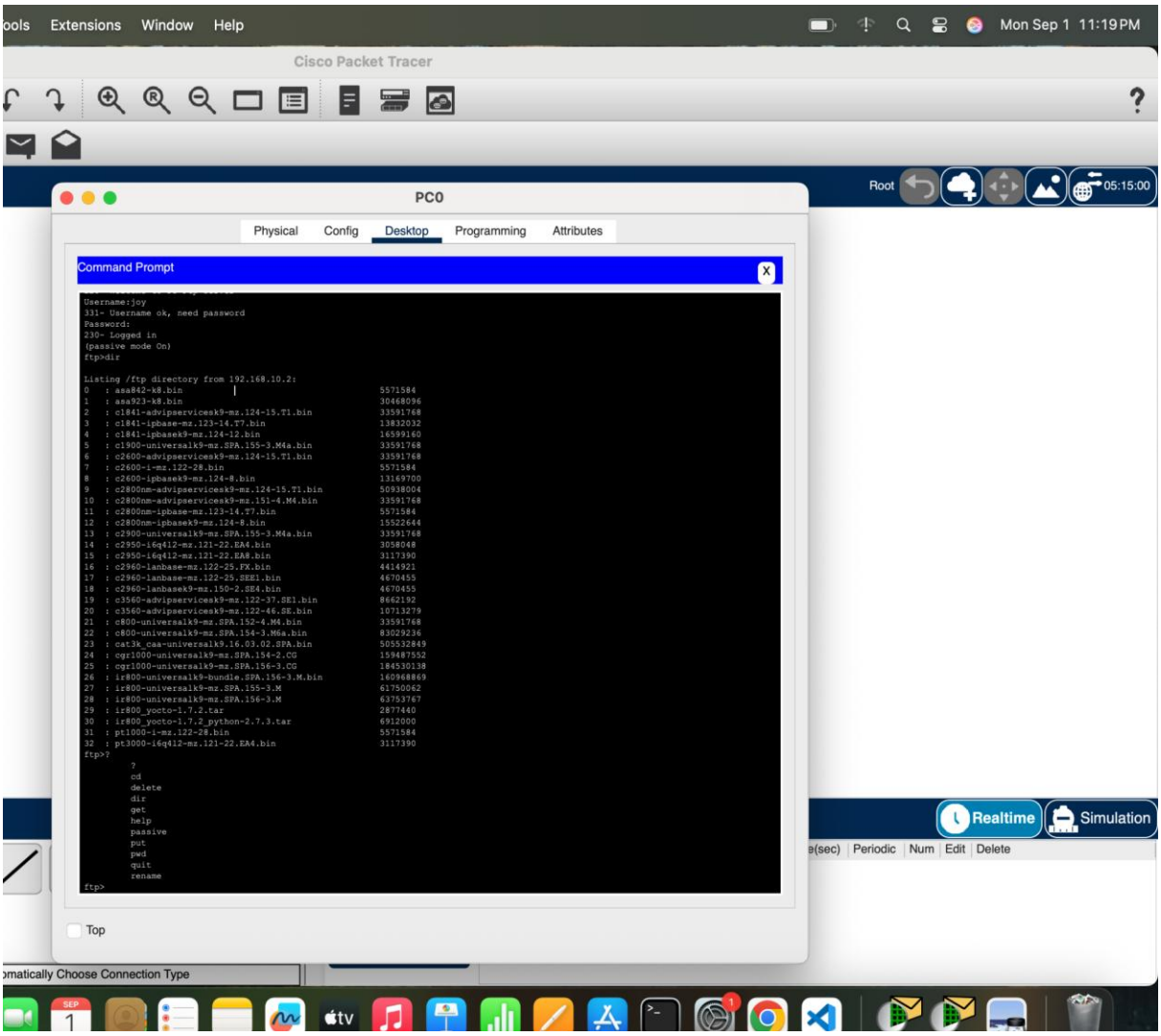


Configure the IP addresses to all devices and add the DNS server's ip to the 'DNS server' section of ip configuration menu.

Lab 4 Implementation of FTP objective
Enable an FTP server to facilitate file transfers

theory
The File Transfer Protocol (FTP) is a standard communication protocol used for the transfer of computer files from a server to a client on a computer network. FTP is built on a client– server model architecture using separate control and data connections between the client and the server. FTP users may authenticate themselves with a clear-text sign-in protocol, normally in the form of a username and password, but can connect anonymously if the server is configured to allow it. For secure transmission that protects the username and password, and encrypts the content, FTP is often secured with SSL/TLS (FTPS) or replaced with SSH File Transfer Protocol (SFTP).

Implementation of FTP server
To implement FTP server, we follow the steps below
Create a network with a server, a switch and end devices.



Assign IP to all devices and server

[Add screenshot here]

Open server and enable FTP service. Create a username and password and assign the r/w privileges and click add.

[Add screenshot here]

Open end device and go to command prompt and type ftp [server ip].use the previously created username and password and log in.

[Add screenshot here]

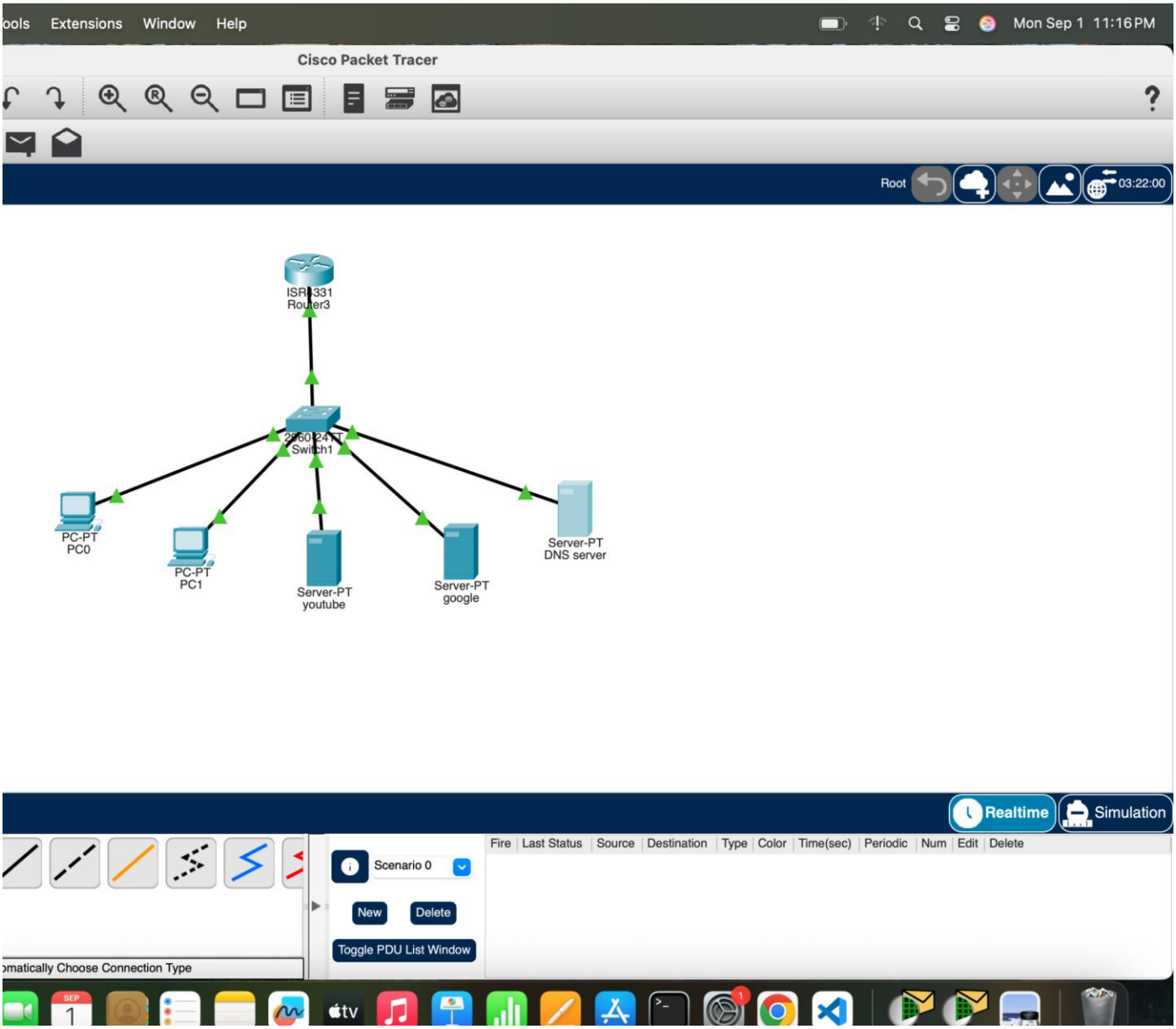
We can now perform file operations on the ftp server using commands like delete, rename etc

Task 4: Implementation of HTTP

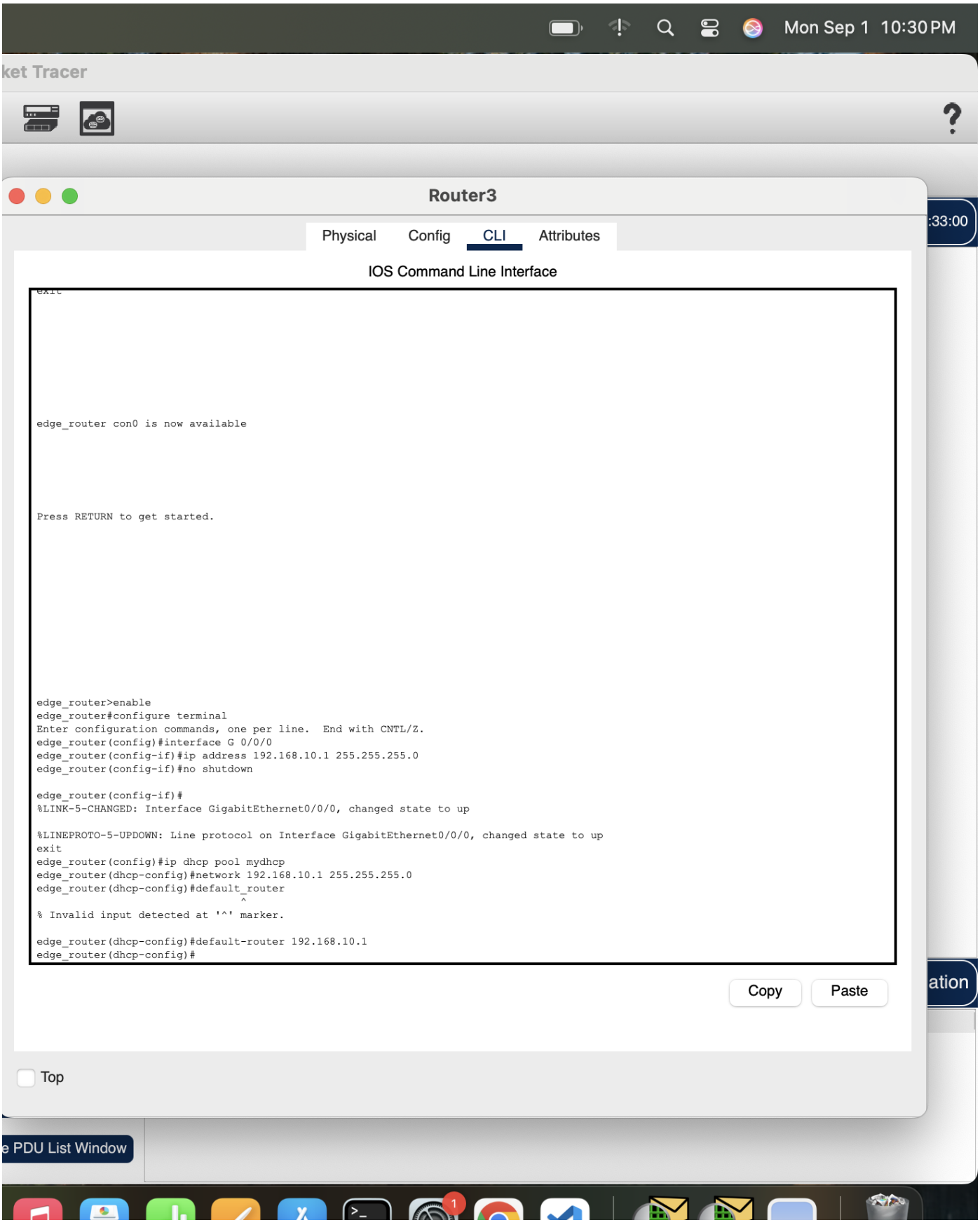
Objective:
To know about the HTTP
To know about the web server, and perform http operation Theory

The implementation of HTTP, or Hypertext Transfer Protocol, involves the creation of software systems that enable communication between clients, such as web browsers, and servers, such as web servers. HTTP operates on a request-response model, where clients send requests to servers, and servers respond with the requested resources or information. The implementation process typically includes developing HTTP servers capable of handling incoming requests, parsing request headers, processing data, and generating appropriate responses. Additionally, HTTP clients are designed to send requests, interpret responses, and handle various status codes. Implementation efforts focus on adhering to HTTP specifications, optimizing performance, ensuring reliability, and incorporating security measures to create robust and efficient systems for web communication.
HTTP working
Http respond to the client request in either of the following two ways:
Sending the file to the client associated with the requested URL.
Generating response by invoking a script and communicating with database.

Configuring a HTTP
To configure a web server, we follow the steps below:
Create a network with a switch, a server and end devices.



Configure IP's of switch server and end devices.



Edit the index.html file and save

Open up browser and type in IP of the webserver and the website will open up.

[Add screenshot here]

Task 5: Implementation of Email Server

Objective: To configure an email server

Theory

An email server implementation involves configuring software systems to manage the sending, receiving, and storage of email messages. This process includes setting up mail transfer agents (MTAs) to route messages between servers, configuring mail delivery agents (MDAs) for storing messages in user mailboxes, and establishing domain name system (DNS) records for proper mail routing. Security measures such as encryption protocols and authentication mechanisms are implemented to protect email communications, while additional features like spam filtering and antivirus scanning enhance reliability and security. Support for email protocols like SMTP, IMAP, and POP3 enables clients to access their emails securely, contributing to the overall goal of providing reliable and efficient email communication services.

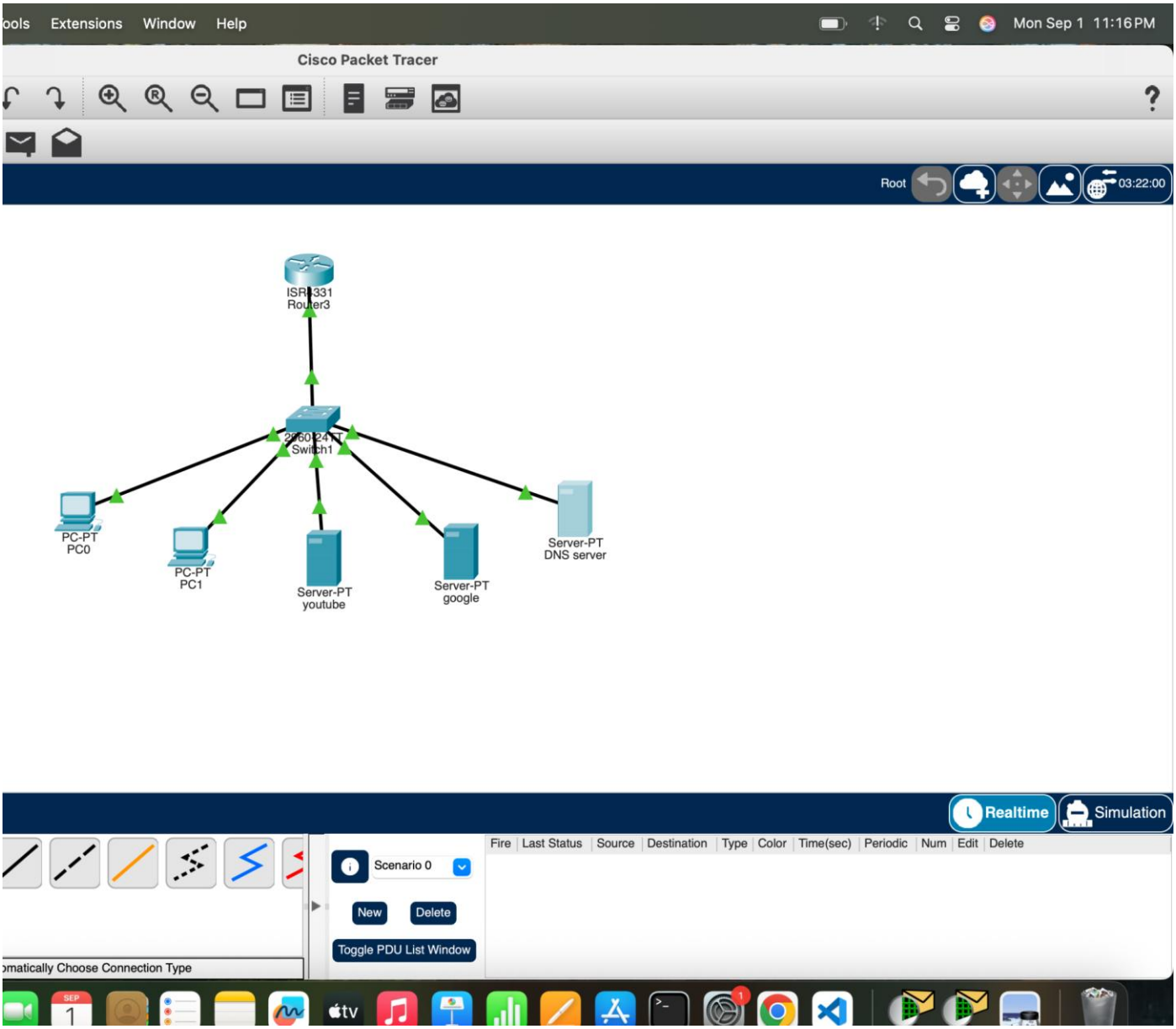
Email server working

Sending the mail in one computer to another computer.

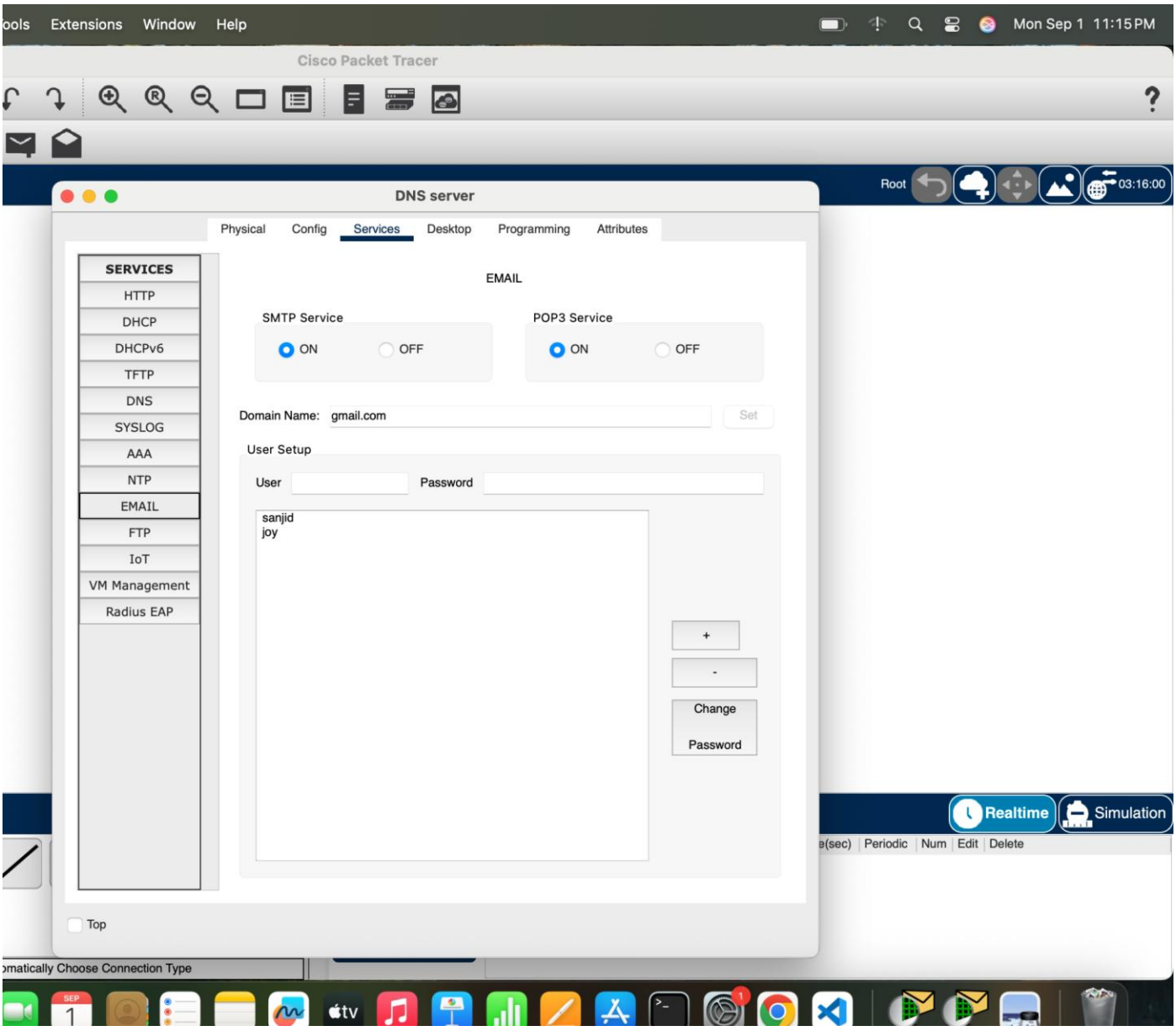
Configuring a Email Server

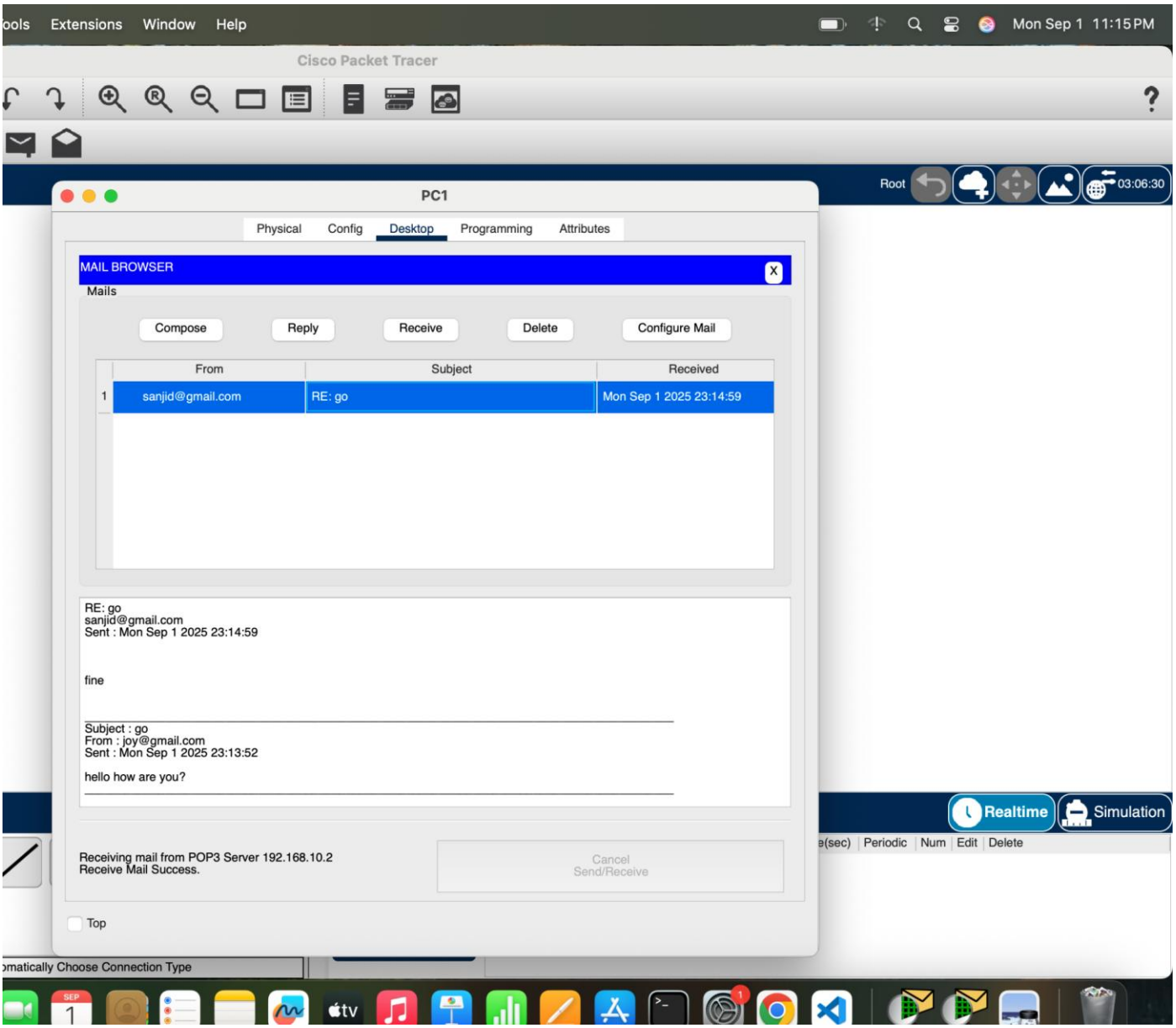
To configure a web server, we follow the steps below:

Create a network with a Switch, a Router, a Server and End Devices

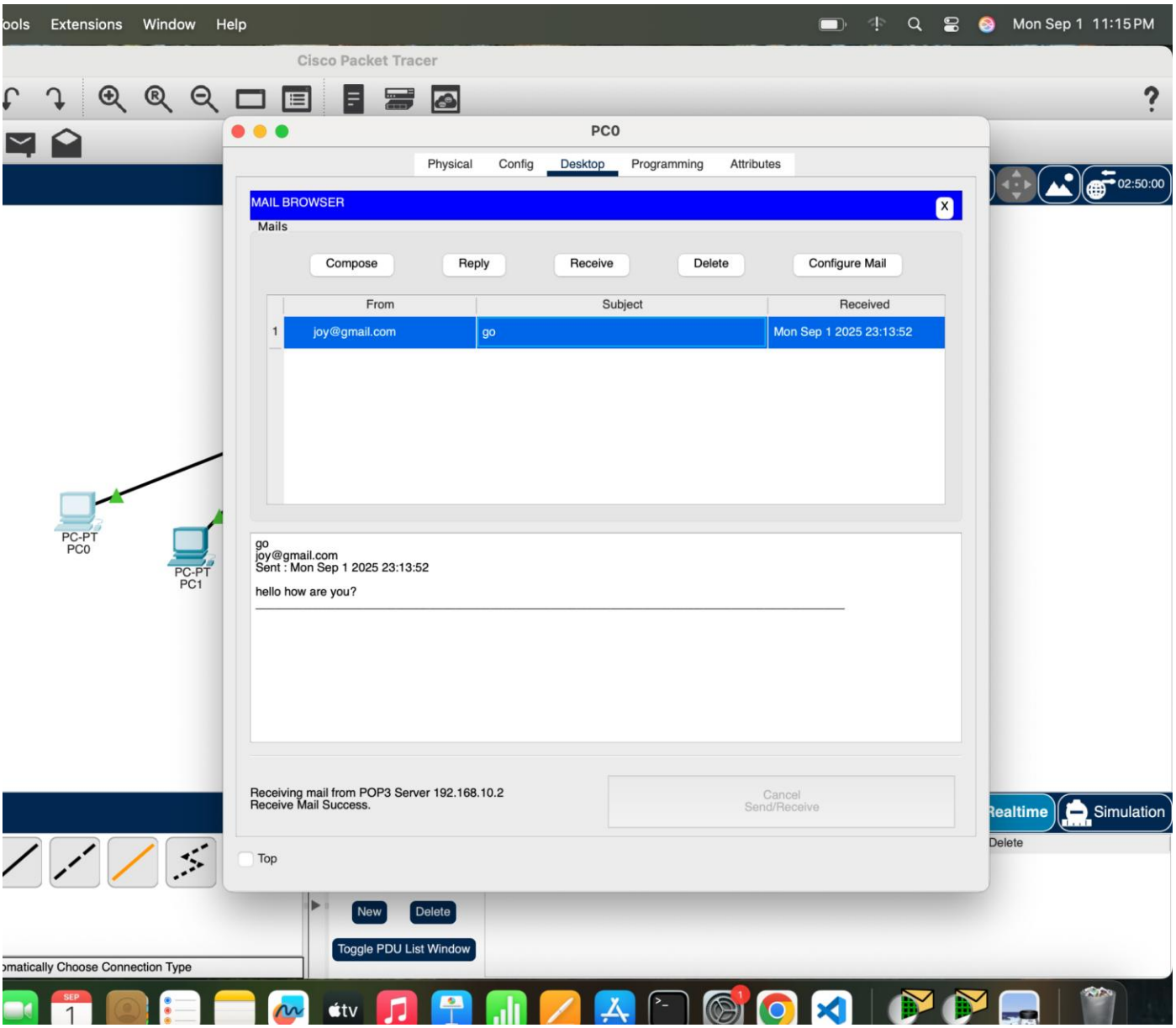


Configure IP's of Switch, Router, Server and End devices.


ools Extensions Window Help
Mon Sep 1 11:15 PM
Cisco Packet Tracer
Root
03:16:00
DNS server
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
EMAIL
SMTP Service
ON
OFF
POP3 Service
ON
OFF
Domain Name: gmail.com
Set
User Setup
User
Password
sanjid
joy
+
-
Change
Password
Top
Realtime
Simulation
(sec)
Periodic
Num
Edit
Delete
matically Choose Connection Type



Open server. Go to services tab. Click Email and turn SMTP and POP3 service on And Fill Create a Domain name as well as user setup.



Open server. Go to services tab. Click Email and configuration the mail.